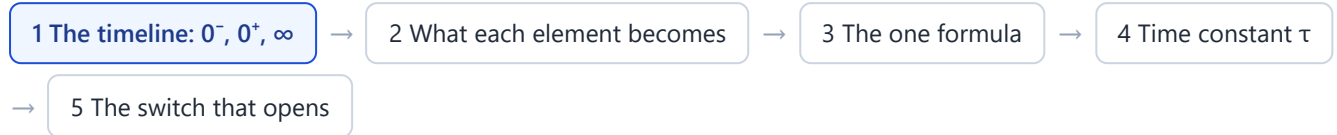


06 — First-order transients

Class notes p.19–23 · 3 board questions · needs 01 (what can't jump), 04 (R_{Th} for the time constant)

Map



The questions this file answers

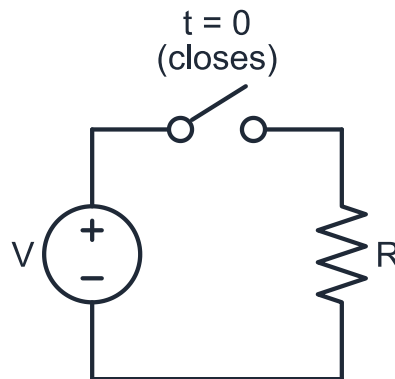
- **Class p.19–20** V , a switch that closes at $t = 0$, R : find I at 0^- , 0^+ , ∞ .
- **Class p.21** V , switch, R , L in series; the switch closes at $t = 0$: find I_R and I_L at 0^- , 0^+ , ∞ .
- **Class p.22–23** 10 V, switch (opens at $t = 0$), 1 k Ω , 1 mH: find I_R and I_L at 0^- and 0^+ .

Build

1 · The timeline

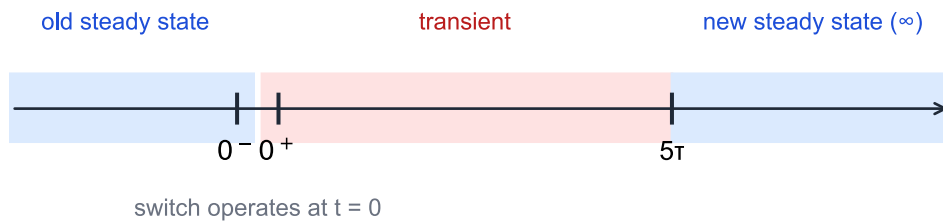
CLASS P.19–20

V , a switch that closes at $t = 0$, and R . Find $I(0^-)$, $I(0^+)$ and $I(\infty)$.



GUESS FIRST

Is there any "transient" at all in this circuit, i.e. any gradual change?



Before the switch: the old steady state. From 0^+ to about 5τ : the transient. After that: the new steady state.

- $t = 0^-$: the instant **just before** the switch operates. Old steady state.
- $t = 0^+$: the instant **just after**.
- $t = \infty$: long after (in practice about 5τ): new steady state.

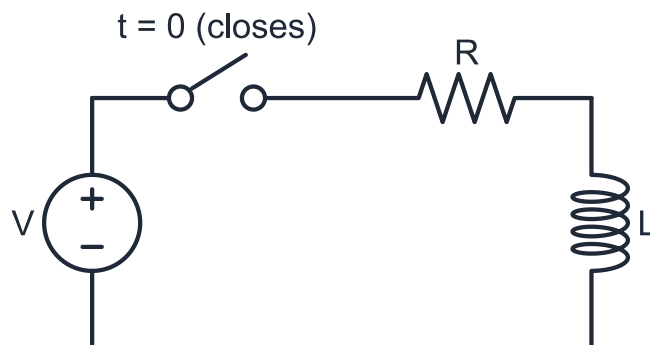
For the resistor-only circuit: $I(0^-) = 0$ (switch open), $I(0^+) = V/R$, $I(\infty) = V/R$. The current jumps straight to its final value, because a resistor has no memory (class p.20: "a resistor allows sudden change").

So the Guess answer: **no transient**. A transient needs an element that stores energy, an L or a C.

2 · What each element becomes at 0^+ and at ∞

CLASS P.21

V , switch, R and L in series; the switch closes at $t = 0$. Find I_R and I_L at 0^- , 0^+ and ∞ .

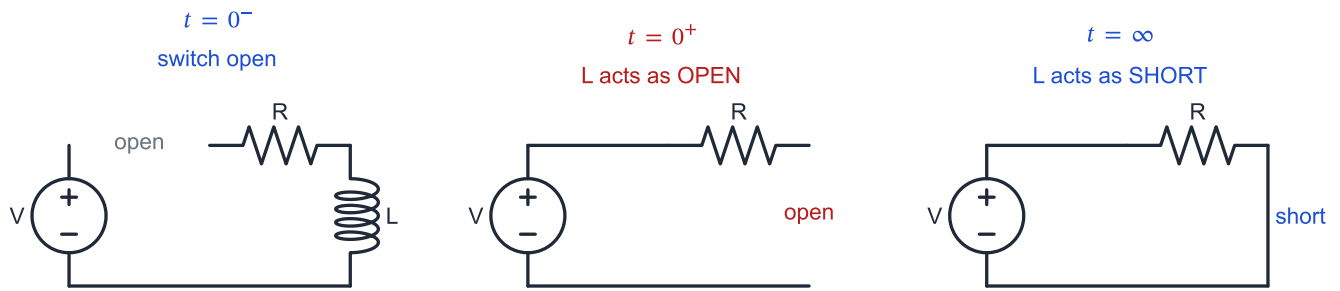


GUESS FIRST

The switch has just closed. The source is pushing. Why can't the current be V/R at $t = 0^+$?

Two rules do all the work (from 01, step 5):

Element	At 0^+ : it keeps its 0^- value	At ∞ (DC steady state)
Inductor	a current source of value $i_L(0^-)$; an open circuit if that value is 0	short circuit
Capacitor	a voltage source of value $v_C(0^-)$; a short circuit if that value is 0	open circuit
Resistor	no memory: whatever the rest of the circuit forces	same



The three snapshots of the class p.21 circuit.

- 0^- : switch open, so $I_R(0^-) = I_L(0^-) = 0$.
- 0^+ : i_L can't jump, so $I_L(0^+) = 0$ and the inductor acts as an **open**. In series, $I_R(0^+) = 0$.
- ∞ : the inductor is a **short**, so $I(\infty) = V/R$.

The current must travel from 0 to V/R . How it travels is step 3.

CHECK 1

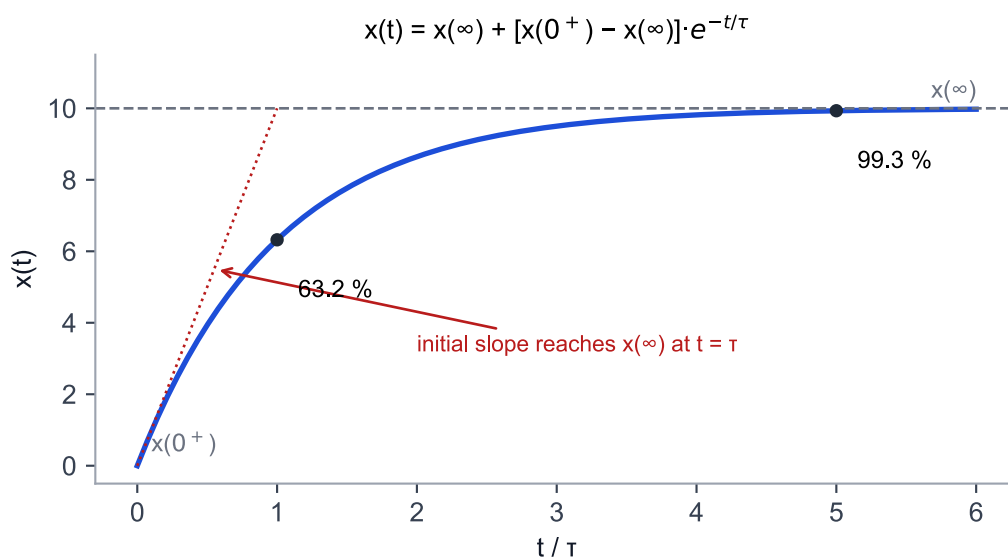
Same circuit with a capacitor instead of the inductor (uncharged at 0^-). Find I at 0^+ and at ∞ .

3 · The one formula for every first-order circuit

Every circuit with **one** L or **one** C (after reduction) moves from its 0^+ value to its ∞ value **exponentially**:

$$x(t) = x(\infty) + [x(0^+) - x(\infty)] \cdot e^{-t/\tau}$$

The class p.22 box: valid for **RL** ✓ and **RC** ✓, not for **RLC** ✗ or **LC** ✗ (those are second order: they can oscillate).



After one τ the change is 63.2 % complete; after 5τ , 99.3 %. The initial slope would reach $x(\infty)$ in exactly one τ .

So a first-order question is always **three numbers**: $x(0^+)$, $x(\infty)$, τ . Then write the formula.

4 · The time constant τ

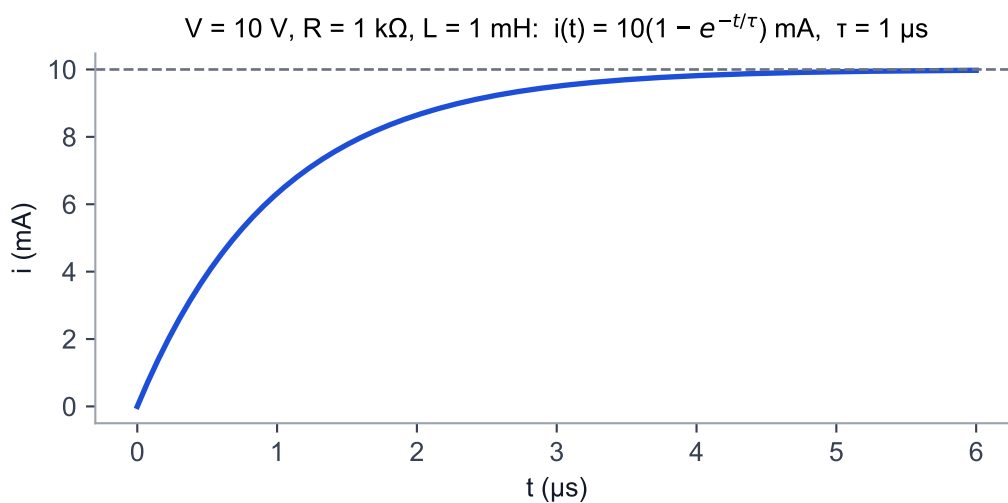
$$\tau = \frac{L}{R} \quad (\text{RL}) \quad \tau = RC \quad (\text{RC})$$

R is the Thévenin resistance seen by the L or C (04), with independent sources switched off. It is the plain R only when the L or C sees a single resistor.

Class p.21 circuit with the p.22 values ($V = 10 \text{ V}$, $R = 1 \text{ k}\Omega$, $L = 1 \text{ mH}$): $x(0^+) = 0$, $x(\infty) = 10 \text{ mA}$, $\tau = \frac{1 \text{ mH}}{1 \text{ k}\Omega} = 1 \text{ }\mu\text{s}$:

μs :

$$i(t) = 10(1 - e^{-t/1 \text{ }\mu\text{s}}) \text{ mA}$$



Confirmed by simulating the circuit step by step: at $t = 3 \text{ }\mu\text{s}$ both give 9.50 mA.

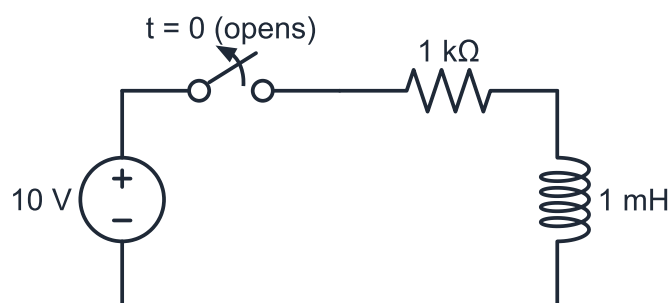
CHECK 2

An RC circuit: $R = 2 \text{ k}\Omega$, $C = 5 \text{ }\mu\text{F}$. Find τ , and the time after which it is "settled" (99 %).

5 · The switch that opens

CLASS P.22-23

10 V, a switch that opens at $t = 0$, $1 \text{ k}\Omega$ and 1 mH in series. Find I_R and I_L at 0^- and 0^+ .



GUESS FIRST

At 0^- the loop is carrying current. At 0^+ the switch is open. What must the inductor do, and where can its current go?

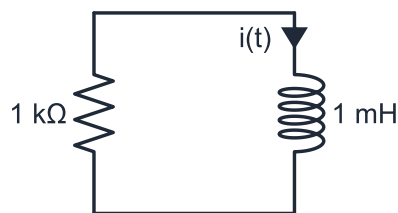
- 0^- : switch closed, steady state, $L = \text{short}$: $I_L(0^-) = 10 \text{ V} / 1 \text{ k}\Omega = 10 \text{ mA} = I_R(0^-)$.
- 0^+ : the inductor current can't jump, so $I_L(0^+) = 10 \text{ mA}$ ✓ (the professor's point).

READ THIS BEFORE WRITING $I_R(0^+) = 0$

The notes give $I_R(0^+) = 0$ alongside $I_L(0^+) = 10 \text{ mA}$. R and L are **in the same series loop**, so they must carry the **same** current. Both statements can't hold. With the switch open there is no path at all for the inductor's 10 mA . In an ideal circuit, $di/dt \rightarrow \infty$ and the inductor voltage spikes. In a real one, the stored energy $\frac{1}{2}Li^2 = \frac{1}{2} \times 1 \text{ mH} \times (10 \text{ mA})^2 = 50 \text{ nJ}$ arcs across the opening contacts.

Ask the professor which path the current takes after $t = 0$. Usually the full question has the switch move the R – L onto its own loop, or has a resistor across the switch.

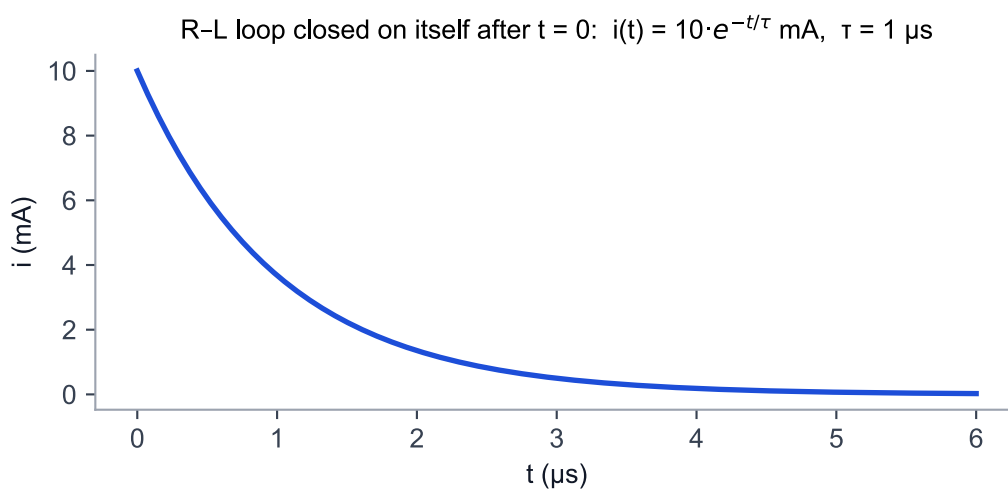
If the R – L loop is closed on itself after $t = 0$ (the usual version of this question):



Illustration, not the board circuit: the R – L loop with the source gone.

$x(0^+) = 10 \text{ mA}$, $x(\infty) = 0$, $\tau = L/R = 1 \mu\text{s}$:

$$i(t) = 10 \cdot e^{-t/1 \mu\text{s}} \text{ mA}$$



Exam form

Procedure for any first-order switching question. 1. $t = 0^-$: old circuit, DC steady state ($L = \text{short}$, $C = \text{open}$). Find $i_L(0^-)$ or $v_C(0^-)$. 2. $t = 0^+$: new circuit. $L \rightarrow \text{current source } i_L(0^-)$; $C \rightarrow \text{voltage source } v_C(0^-)$. Find the asked quantity. 3. $t = \infty$: new circuit, DC steady state ($L = \text{short}$, $C = \text{open}$). 4. $\tau = L/R_{Th}$ or $R_{Th}C$, with R_{Th} seen by the L or C . 5. $x(t) = x(\infty) + [x(0^+) - x(\infty)]e^{-t/\tau}$.

Facts to state. Inductor current and capacitor voltage cannot change instantaneously. Settling $\approx 5\tau$ (99.3 %). 63.2 % of the change happens in one τ .

Traps

- Using the ∞ circuit at 0^+ (treating L as a short immediately).
- Calling $i_L(0^+) = 0$ when the inductor was carrying current at 0^- : it keeps its 0^- value, whatever it was.
- $\tau = L/R$ with the wrong R : it must be the Thévenin resistance the L sees.
- Using the first-order formula on an RLC circuit.

Self-test

1. What does a capacitor look like at 0^+ if it was uncharged? And at ∞ ?
2. Why is $\tau = L/R$ and not LR ? Check with units: $H/\Omega = ?$
3. RL circuit, $i(0^+) = 2 \text{ A}$, $i(\infty) = 0$, $\tau = 5 \text{ ms}$. Write $i(t)$. What is i at 10 ms ?
4. After how many time constants is a transient 99 % done?
5. Why is there no transient in a purely resistive circuit?

Answers

Check 1. Uncharged C = short at 0^+ , so $i(0^+) = \mathbf{V/R}$. At ∞ the C is open, so $i(\infty) = \mathbf{0}$. (The mirror image of the inductor.)

Check 2. $\tau = 2 \text{ k}\Omega \times 5 \text{ }\mu\text{F} = \mathbf{10 \text{ ms}}$; settled after about $5\tau = \mathbf{50 \text{ ms}}$.

Self-test.

1. A **short** at 0^+ ; an **open** at ∞ .
2. $H/\Omega = (V \cdot s/A)/(V/A) = \mathbf{s}$: L/R is a time. (LR has units $\Omega^2 \cdot s$, not a time.)
3. $i(t) = \mathbf{2e^{-t/5 \text{ ms}} \text{ A}}$; at 10 ms , $i = 2e^{-2} = \mathbf{0.27 \text{ A}}$.
4. About $\mathbf{5\tau}$ (99.3 %); 4.6τ gives exactly 99 %.
5. A resistor stores no energy, so nothing needs time to build up or run down: every quantity jumps straight to its final value.